In [2]:
```
import pandas as pd
import numpy as np
df=pd.DataFrame(np.random.randn(4,3), index=['a','c','d','f'],
columns=['one','two', 'three'])
df.reindex(['a','b','c','d','e','f'])
print("Original Dataframe with NaN\n",df)
print("\n")
print("Dropped DataFrame\n",df.dropna())
print("\n")
```

```
Original Dataframe with NaN
         one       two     three
a -0.104998 -0.771593 -1.076597
c  1.226158 -0.999080  0.500628
d -0.642702 -0.291846  0.803269
f  1.933353  0.238156 -0.947403


Dropped DataFrame
         one       two     three
a -0.104998 -0.771593 -1.076597
c  1.226158 -0.999080  0.500628
d -0.642702 -0.291846  0.803269
f  1.933353  0.238156 -0.947403
```

In [5]:
```
df=pd.DataFrame(np.random.randn(4,3), index=['a','c','d','f'],
columns=['one','two', 'three'])
df.reindex(['a','b','c'])
print("Original Dataframe:\n",df)
print("\n")
print("NaN replaced with '0':\n")
print(df.fillna(0))
```

```
Original Dataframe:
         one       two     three
a  1.256180 -0.782115 -0.418294
c  0.216061 -0.081155 -0.917034
d -1.693934  0.753946 -0.131457
f  0.293285 -1.289652 -1.361051


NaN replaced with '0':

        one       two     three
a  1.256180 -0.782115 -0.418294
c  0.216061 -0.081155 -0.917034
d -1.693934  0.753946 -0.131457
f  0.293285 -1.289652 -1.361051
```

In [7]:
```
df=pd.DataFrame(np.random.randn(4,3), index=['a','c', 'd','f'],
columns=['one','two', 'three'])
df.reindex(['a','b','c'])
print("Original Dataframe:\n",df)
print("\n")
print("NaN replaced with '0':\n")
print(df.fillna(0))
```

```
Original Dataframe:
         one       two     three
a -0.196615  0.856486  1.783978
c -0.419945 -1.127086  0.631272
d -0.526850  2.277711 -2.388132
f -1.070542  0.212567  1.007895


NaN replaced with '0':

        one       two     three
a -0.196615  0.856486  1.783978
c -0.419945 -1.127086  0.631272
d -0.526850  2.277711 -2.388132
f -1.070542  0.212567  1.007895
```

In [8]:
```
df-pd.DataFrame({'one': [10,20,80,40,50],
'two': [60,70,80,0,10]})
print("Original Dataframe:\n",df)
print("\n DataFrame with replaced values:\n")
print(df.replace({10:5,80:3}))
```

```
Original Dataframe:
         one       two     three
a -0.196615  0.856486  1.783978
c -0.419945 -1.127086  0.631272
d -0.526850  2.277711 -2.388132
f -1.070542  0.212567  1.007895

 DataFrame with replaced values:

        one       two     three
a -0.196615  0.856486  1.783978
c -0.419945 -1.127086  0.631272
d -0.526850  2.277711 -2.388132
f -1.070542  0.212567  1.007895
```

In [10]:
```
df=pd.DataFrame(np.random.randn(4,3), index=['a','c','d','f'],
columns=['one', 'two', 'three'])
df=df.reindex(['a','b','e','d','e','f'])
print("Original Dataframe:\n",df)
print("\nNan-with True fill:\n")
print(df['one'].isnull())
```

```
Original Dataframe:
         one       two     three
a -0.092824 -1.480851  1.076070
b       NaN       NaN       NaN
e       NaN       NaN       NaN
d -1.736752  1.780538 -0.917044
e       NaN       NaN       NaN
f  1.083376  1.954798  0.587135

Nan-with True fill:

a    False
b     True
e     True
d    False
e     True
f    False
Name: one, dtype: bool
```

In [12]:
```
df=pd.DataFrame(np.random.randn(4,3), index=['a','c','d','f'],
columns=['one', 'two', 'three'])
df=df.reindex(['a','b','c','d','e','f'])
print("Original Dataframe:\n",df)
print("\nNaN - Filled with Background values:\n")
print(df.fillna(method='bfill'))
```

```
Original Dataframe:
         one       two     three
a -0.296933 -1.746561 -1.872767
b       NaN       NaN       NaN
c  2.098283  1.604779  0.907239
d  0.394074  0.324438  0.233618
e       NaN       NaN       NaN
f  0.717836  0.844929  0.213024

NaN - Filled with Background values:

        one       two     three
a -0.296933 -1.746561 -1.872767
b  2.098283  1.604779  0.907239
c  2.098283  1.604779  0.907239
d  0.394074  0.324438  0.233618
e  0.717836  0.844929  0.213024
f  0.717836  0.844929  0.213024
```

In [ ]:

```

```